EOP-LLM: Energy Oriented Pruning for Large Language Models

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Introduction

In recent years, large language models (LLMs) have been widely deployed in many real-world systems. As LLM deployment continues to scale, inference energy consumption is growing rapidly and has become a key bottleneck for large-scale deployment. In practical deployment environments, such as AI data centers and edge devices, the available energy headroom for inference can fluctuate due to power caps, co-located workloads, thermal constraints, and device-level power limits. Meanwhile, different requests vary significantly in sequence length and computational demand, making request-level energy control necessary for practical LLM inference. However, most existing LLM inference-efficiency methods mainly focus on reducing FLOPs or latency through static compression and dynamic computation, rather than explicitly modeling or enforcing real device energy consumption under per-sequence energy budgets.

Key Contributions

- We propose EOP-LLM: an energy-oriented dynamic pruning framework that enables LLM inference under explicit per-sequence energy budgets while keeping the pretrained backbone frozen.
- We design a device-calibrated energy model: a joule-level energy proxy that decomposes inference energy into attention-side memory traffic, attention-side computation, FFN-side memory traffic, and FFN-side computation, with coefficients fitted from short on-device power measurement sweeps.
- We develop joint token and FFN energy-aware execution: lightweight token and FFN selectors identify important token blocks and FFN channel blocks, while a global budget coordinator converts their scores into energy-feasible execution decisions.

Related Work

Static Compression Methods.

Methods such as GPTQ, LLM-Pruner, LoRAPrune, MINITRON, SliM-LLM, and SlimLLM reduce inference cost by compressing the pretrained LLM before deployment through quantization, pruning, or knowledge distillation. However, the resulting model is fixed after compression, so its execution behavior cannot dynamically adapt to different input requests or changing per-sequence energy budgets.

Dynamic Computation Methods.

Methods such as Dynamic Context Pruning (DCP), LazyLLM, Probe Pruning, SlimInfer, SkipGPT, and DiffSkip reduce inference-time computation by pruning tokens or KV-cache entries, skipping layers, or dynamically gating attention and FFN components. However, these methods primarily optimize FLOPs or latency rather than real device energy, so they cannot reliably enforce per-sequence energy-budget satisfaction.

Key Challenges

The three key challenges to enabling LLM inference under explicit per-sequence energy budgets are:

- **Device-Calibrated Energy Modeling.**

  Real inference energy is not determined by FLOPs alone. It also depends on memory traffic, kernel execution behavior, hardware overhead, and the target device. How can we accurately estimate the joule-level energy cost of different token-retention and FFN-width execution choices on a given device?

- **Per-Sequence Budget Enforcement.**

  During autoregressive inference, energy is consumed stage by stage, while each request must remain within its own per-sequence budget. How can the model track the remaining energy budget during generation and convert soft importance scores into hard execution decisions without violating the budget?

- **Joint Token-FFN Computation Allocation.**

  Token pruning reduces attention and KV-cache costs, while FFN width selection reduces feed-forward computation. These two decisions compete for the same energy budget. How can we jointly allocate energy between token retention and FFN execution to preserve model quality under tight energy constraints?

Methodology

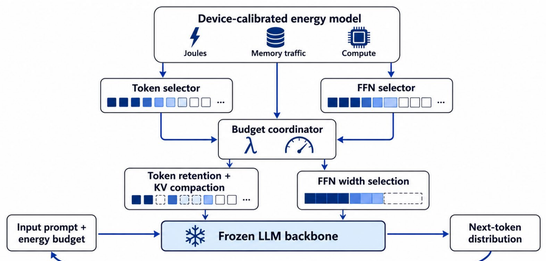


Fig. 1 Architecture Overview

1. Device-Calibrated Energy Model

- **Energy Proxy.** Estimate the joule cost of an execution policy:

$$\hat{E}_q(x,y;\pi)=\sum_{t=0}^{T}\sum_{\ell=1}^{L}\hat{E}_{q,\ell,t}(n_t,n_{\ell,t}^{keep},w_{\ell,t}^{keep})+\gamma_q,$$

  where $n_{\ell,t}^{keep}$ is the retained token length, $w_{\ell,t}^{keep}$ is the selected FFN width, and γ_q is the overhead.

- **Energy Decomposition.** To capture real device energy beyond FLOPs, each layer-stage cost is decomposed into memory and compute terms:

$$\hat{E}_q=\sum_{t,\ell}\left(\alpha_{q,\ell}^{HBM}\mathrm{Bytes}_{\ell,t}^{att}+\alpha_{q,\ell}^{MAC}\mathrm{MAC}_{\ell,t}^{att}+\beta_{q,\ell}^{HBM}\mathrm{Bytes}_{\ell,t}^{ffn}+\beta_{q,\ell}^{MAC}\mathrm{MAC}_{\ell,t}^{ffn}\right)+\gamma_q,$$

  where $\mathrm{Bytes}_{\ell,t}^{att/ffn}$ and $\mathrm{MAC}_{\ell,t}^{att/ffn}$ denote memory traffic and computation cost, $\alpha_{q,\ell}^{HBM},\alpha_{q,\ell}^{MAC},\beta_{q,\ell}^{HBM},\beta_{q,\ell}^{MAC}$. are device−calibrated coefficients.

- **Calibration.** Device coefficients are fitted via least-squares regression using runtime power measurement sweeps collected from NVML and RAPL telemetry.

2. Token and FFN Selectors

- **Token Selector.** Scores each token block using hidden-state statistics, position information, activation history, and local uncertainty:

$$\rho_{b,t}=[\mathrm{mean}_{i\in I_{b,t}}H_{t,i}^{(\ell_0)},\max_{i\in I_{b,t}}H_{t,i}^{(\ell_0)},\mathrm{std}_{i\in I_{b,t}}H_{t,i}^{(\ell_0)},h_{b,t},\mathrm{pos}_{b,t},\omega_{b,t}],$$

$$s_{\ell,t}=P_t^{(2)}\nu_\ell+\iota_\ell\mathbf{1},\ell\in[L],$$

  where $\mathrm{pos}_{b,t}$ encodes block position, $\omega_{b,t}$ is a detached EMA of block-aligned activations, and $h_{b,t}$ captures local uncertainty. ν_ℓ,ι_ℓ are learned per-layer parameters.

- **FFN Selector.** Scores each FFN channel block using static weight information, activation usage, and a Fisher-style gradient signal:

$$r_{\ell,j,t}=[\|W_2^{(\ell)}[:,K_{\ell,j}]\|_F,\|W_1^{(\ell)}[K_{\ell,j},:]\|_F,\mathrm{EMA}_t(\|\xi_{\ell,j,t}\|_2),\mathrm{EMA}_t(\|\nabla_{\xi_{\ell,j,t}}\mathcal{L}\|_2)],$$

$$z_{\ell,j,t}=\eta_\ell^\top r_{\ell,j,t},$$

  where $W_1^{(\ell)}$ and $W_2^{(\ell)}$ are FFN projection matrices, $\|\cdot\|_F$ is the Frobenius norm, $\xi_{\ell,j,t}$ is the FFN intermediate activation restricted to $K_{\ell,j}$,and $\mathcal{L}$ is the loss.

3. Energy-Priced Budget Coordination

- **Remaining Energy Budget.** Updates remaining budget stage by stage during autoregressive inference:

$$E_0^{rem}=E^{bud}(x,T)-\gamma_q,\qquad E_{t+1}^{rem}=\left[E_t^{rem}-\hat{E}_q^{(t)}(x,y;\pi)\right]_+,$$

  where $E^{bud}(x,T)$ is the per-sequence energy budget, γ_q is the device-dependent constant overhead, E_t^{rem} is the remaining budget at stage t, and $\hat{E}_q^{(t)}$ represents stage-t proxy energy.

- **Energy-Priced Token Gates.** Converts token-block importance scores into energy-aware token gates:

$$\Delta_{\ell,t}^{tok}=\frac{\partial\hat{E}_{q,\ell,t}^{att}}{\partial\bar{n}_{\ell,t}},\qquad \tilde{s}_{\ell,b,t}=s_{\ell,b,t}-\lambda\mid I_{b,t}\mid\Delta_{\ell,t}^{tok},\qquad g_{\ell,b,t}\in[0,1],$$

  where $s_{\ell,b,t}$ is the token-block importance score, $I_{b,t}$ is the token block, $\Delta_{\ell,t}^{tok}$ is the marginal attention energy cost, and λ denotes the global energy price.

- **Energy-Priced FFN Gates.** Allocates the residual budget after attention to FFN channel-block execution:

$$E_t^{ffn}=[E_t^{rem}-\hat{E}_{q,t}^{att}]_+,\qquad \tilde{z}_{\ell,j,t}=\sigma(z_{\ell,j,t}-\lambda e_{\ell,j,t}),\qquad m_t=\arg\min_{0\le m\le 1,\ e_t^\top m\le E_t^{ffn}}\frac{1}{2}\|m-\tilde{z}_t\|_2^2,$$

  Where $z_{\ell,j,t}$ is the FFN-block score, $e_{\ell,j,t}$ is the marginal FFN energy cost, E_t^{ffn} is the residual FFN energy budget, and m_t is the relaxed FFN execution gates.

4. Execution Policy Training and Hardening

- **Training Objective.** Trains only the lightweight selector parameters under a single per-sequence energy contract:

$$\min_\phi\max_{\lambda\ge 0}\mathbb{E}_{(x,y)\sim\mathcal{D}}\left[-\sum_{t=1}^{T}\log p_{\theta,\pi_\phi}(y_t|x,y<t)+\lambda\left(\hat{E}_q(x,y;\pi_\phi)-E^{bud}(x,T)\right)\right],$$

$$\lambda\leftarrow\left[\lambda+\eta_\lambda\left(\hat{E}_q(x,y;\pi_\phi)-E^{bud}(x,T)\right)\right]_+,$$

  where θ is the frozen pretrained LLM, ϕ is the selector parameters, and λ is the global dual price.

- **Inference-Time Hardening.** Converts relaxed gates into binary execution decisions that satisfy the energy budget:

$$\hat{E}_{q,t}^{att}\left(\{\hat{\mathcal{B}}_{\ell,t}\}_{\ell=1}^{L}\right)\le E_{q,t}^{att,rlx},\quad \hat{g}_{\ell,b,t}=\mathbf{1}\{b\in\hat{\mathcal{B}}_{\ell,t}\},$$
$$\hat{m}_t\in\{0,1\}^{d_m},\quad e_t^\top\hat{m}_t\le E_t^{ffn},$$

  where $\hat{g}_{\ell,b,t}$ and $\hat{m}_t$ are binary token and FFN execution gates, $E_{q,t}^{att,rlx}$ is the relaxed attention budget, and E_t^{ffn} is the residual FFN budget.

- **Execution Sets.** Executes only the selected token positions and selected FFN channels:

$$S_{\ell,t}=\bigcup_{b:\hat{g}_{\ell,b,t}=1}I_{b,t},\qquad C_{\ell,t}=\bigcup_{j:\hat{m}_{\ell,t}=1}K_{\ell,j},$$

  where $S_{\ell,t}$ is the retained token positions, and $C_{\ell,t}$ is the selected FFN channels.

Experimental Results

A. Energy Model Calibration

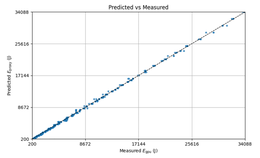


Figure 1: Overall energy calibration: predicted versus measured energy (LLaMA 3.2–3B).

Main Findings

- Energy proxy closely tracks real device energy.
- $R^2 = 0.9982$ over 3,071 calibration samples.
- Weighted relative error: 1.99%.

B. Experimental Results

Model	Method	Ψ = 0.5 PPL	Δ%	Ψ = 0.6 PPL	Δ%	Ψ = 0.7 PPL	Δ%	Ψ = 0.8 PPL	Δ%	Ψ = 1.0 PPL	Δ%
LLaMA 3.2-1B	Dense	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	16.35	0
	DCP	550.74	-1.23	220.43	-1.02	32.33	-0.82	22.55	-0.61	16.35	0
	PP	133.51	-1.58	85.71	-1.35	26.95	-0.77	19.85	-0.58	16.35	0
	Ours (token-only)	510.32	0.39	203.43	0.47	30.14	0.34	20.55	0.32	16.35	0
	Ours (FFN-only)	121.76	0.41	73.44	0.52	24.01	0.49	18.49	0.51	16.35	0
	Ours	**38.07**	**0.22**	**20.29**	**0.23**	**17.75**	**0.19**	**16.93**	**0.17**	16.35	0
LLaMA 3.2-3B	Dense	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13.12	0
	DCP	292.21	-1.21	171.11	-1.23	26.32	-1.02	17.61	-0.71	13.12	0
	PP	91.78	-1.47	53.45	-1.35	22.31	-0.97	15.12	-1.38	13.12	0
	Ours (token-only)	270.76	0.49	157.91	0.45	24.54	0.32	16.05	0.35	13.12	0
	Ours (FFN-only)	87.48	0.42	49.8	0.48	20.18	0.24	14.78	0.31	13.12	0
	Ours	**21.26**	**0.21**	**15.75**	**0.22**	**14.11**	**0.17**	**13.22**	**0.16**	13.12	0
LLaMA 3.1-8B	Dense	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10.73	0
	DCP	266.92	-1.21	140.62	-1.13	22.78	-0.92	14.41	-0.66	10.73	0
	PP	75.74	-1.47	47.95	-1.25	18.58	-0.87	13.05	-0.98	10.73	0
	Ours (token-only)	247.33	0.49	137.16	0.46	21.24	0.33	13.14	0.34	10.73	0
	Ours (FFN-only)	69.82	0.42	43.12	0.35	17.33	0.17	12.73	0.24	10.73	0
	Ours	**19.42**	**0.19**	**13.68**	**0.16**	**12.21**	**0.11**	**10.82**	**0.09**	10.73	0

Table 1: Average PPL over 3 trials on WikiText-2 (raw) at target ratios Ψ.

Model	Method	Ψ	Δ%	BoolQ	PIQA	HellaSwag	WinoG	ARC-c	ARC-e	OBQA	Avg.
LLaMA 3.1-8B	Dense	1.00	0.00	80.28	78.24	76.18	72.19	46.49	82.59	43.20	65.45
	DCP	0.60	-1.13	62.75	64.64	56.27	55.72	27.09	63.13	28.60	47.77
	PP	0.60	-1.25	72.72	73.88	66.12	65.11	34.11	74.07	34.40	57.20
	Ours (token-only)	0.60	0.46	63.33	65.78	56.28	53.12	27.76	44.44	28.80	48.50
	Ours (FFN-only)	0.60	0.35	74.98	74.43	66.77	65.51	35.79	74.81	34.60	58.13
	Ours	**0.60**	**0.19**	**78.29**	**77.86**	**72.97**	**69.06**	**42.14**	**79.26**	**40.20**	**62.83**
	DCP	0.50	-1.21	55.47	56.91	43.38	47.36	23.08	35.19	23.40	40.68
	PP	0.50	-1.47	66.55	70.13	61.12	59.98	30.43	50.37	30.60	53.02
	Ours (token-only)	0.50	0.49	56.36	57.13	44.38	47.75	23.41	37.78	23.80	41.52
	Ours (FFN-only)	0.50	0.42	69.82	71.44	62.06	60.22	30.77	50.74	31.20	53.75
	Ours	**0.50**	**0.22**	**74.34**	**75.14**	**68.38**	**65.98**	**38.13**	**56.67**	**36.80**	**59.35**

Table 2: Average commonsense accuracy % over 3 trials on seven benchmarks for LLaMA 3.1-8B

Model	Method	Ψ	Δ%	BoolQ	PIQA	HellaSwag	WinoG	ARC-c	ARC-e	OBQA	Avg.
LLaMA 3.2-1B	Dense	1.00	0.00	57.77	72.96	55.54	54.38	31.44	49.63	29.80	50.22
	DCP	0.60	-1.02	43.46	48.48	39.93	41.12	19.06	29.26	18.60	34.27
	PP	0.60	-1.35	51.96	65.94	46.81	48.46	24.75	40.37	23.40	43.10
	Ours (token-only)	0.60	0.47	45.41	50.87	40.43	42.62	20.74	31.48	19.60	35.88
	Ours (FFN-only)	0.60	0.52	52.45	67.57	47.92	49.49	25.08	41.48	23.80	43.97
	Ours	0.60	**0.21**	**55.93**	**70.13**	**52.47**	**54.14**	**29.43**	**47.04**	**27.60**	**48.11**
LLaMA 3.2-3B	DCP	0.50	-1.23	38.72	44.18	32.33	38.75	16.72	27.04	16.20	30.56
	PP	0.50	-1.58	47.89	60.99	38.52	47.43	22.07	34.07	20.20	38.74
	Ours (token-only)	0.50	0.48	38.96	45.81	32.81	39.07	17.06	27.78	16.40	31.13
	Ours (FFN-only)	0.50	0.34	49.36	63.49	39.45	47.91	22.41	35.56	21.40	39.94
	Ours	0.50	**0.22**	**54.16**	**69.48**	**49.11**	**53.35**	**27.42**	**44.81**	**24.80**	**46.16**

Table 3: Average commonsense accuracy % over 3 trials on seven benchmarks for LLaMA 3.1-1B.

Model	Method	Ψ	Δ%	BoolQ	PIQA	HellaSwag	WinoG	ARC-c	ARC-e	OBQA	Avg.
LLaMA 3.2-3B	Dense	1.00	0.00	68.81	75.19	65.57	63.69	39.13	58.15	36.80	58.19
	DCP	0.60	-1.23	53.82	60.23	49.17	48.35	23.08	38.15	24.20	42.40
	PP	0.60	-1.15	61.16	71.93	59.81	58.25	31.44	50.37	30.60	51.94
	Ours (token-only)	0.60	0.45	54.83	61.70	50.33	49.41	24.48	39.26	24.80	43.54
	Ours (FFN-only)	0.60	0.48	62.91	72.03	60.67	59.75	32.78	51.85	31.80	53.11
	Ours	0.60	**0.21**	**66.36**	**74.21**	**62.72**	**63.06**	**35.45**	**55.56**	**34.20**	**55.94**
LLaMA 3.1-8B	DCP	0.50	-1.21	49.97	52.71	41.14	45.54	21.41	34.22	20.80	37.97
	PP	0.50	-1.47	58.13	66.76	53.34	55.41	27.09	45.56	27.20	47.64
	Ours (token-only)	0.50	0.49	51.68	53.68	42.69	46.25	21.75	34.97	21.20	38.86
	Ours (FFN-only)	0.50	0.42	59.76	67.74	55.32	57.22	27.76	46.67	28.20	48.95
	Ours	0.50	**0.23**	**63.12**	**73.01**	**59.87**	**61.72**	**30.77**	**52.22**	**30.20**	**52.99**

Table 4: Average commonsense accuracy % over 3 trials on seven benchmarks for LLaMA 3.1-3B.